2.38 Integration: Surface Area

Worked Solutions

1 a)

$$y = 3x + 5, \qquad \frac{dy}{dx} = 3$$

$$1 + \frac{dy}{dx}^2 = 1 + (3)^2 = 10, \qquad \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{10}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = (3x + 5)\sqrt{10}$$

$$S = 2\pi \int_0^2 y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = 2\sqrt{10}\pi \int_0^2 (3x + 5)\, dx$$

$$= 2\sqrt{10}\pi \left[\frac{3}{2}x^2 + 5x\right]_0^2$$

$$= 2\sqrt{10}\pi[6 + 10 - 0] = 32\sqrt{10}\pi \text{ units}^2$$

1 b)

$$y = 2\sqrt{x} = 2x^{1/2}, \qquad \frac{dy}{dx} = x^{-1/2}$$

$$1 + \frac{dy}{dx}^2 = 1 + (x^{-1/2})^2 = 1 + \frac{1}{x}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{1}{x}}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 2\sqrt{x}\sqrt{1 + \frac{1}{x}} = 2\sqrt{x}\sqrt{\frac{x+1}{x}} = 2\sqrt{x+1}$$

$$S = 2\pi \int_0^3 y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = 2\pi \int_0^3 2(x+1)^{1/2}\, dx$$

$$= 2\pi \left[\frac{4}{3}(x+1)^{3/2}\right]_0^3$$

$$= 2\pi \left[\frac{32}{3} - \frac{4}{3}\right] = \frac{56\pi}{3} \text{ units}^2$$

1 c)

$$y = \cosh x, \qquad \frac{dy}{dx} = \sinh x$$

$$1 + \frac{dy}{dx}^2 = 1 + \sinh^2 x = \cosh^2 x$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\cosh^2 x} = \cosh x$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \cosh x \cosh x = \cosh^2 x = \frac{1}{2} + \frac{1}{2}\cosh 2x$$

$$
\begin{aligned}
S &= 2\pi \int_0^1 y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi\int_0^1 \left(\frac{1}{2}+\frac{1}{2}\cosh 2x\right)\,dx \\
&= \pi\int_0^1 (1+\cosh 2x)\,dx = \pi\left[x+\frac{1}{2}\sinh 2x\right]_0^1 \\
&= \pi\left(1+\frac{1}{2}\sinh 2\right) = \frac{\pi}{2}(2+\sinh 2) \\
&= \frac{\pi}{2}\left(2+\frac{1}{2}e^2-\frac{1}{2}e^{-2}\right) = \frac{\pi}{4}(4+e^2-e^{-2})\ \text{units}^2
\end{aligned}
$$

1 d)

$$
\begin{aligned}
y &= \frac{1}{2\sqrt{3}}x^{1/2}(4-x) = \frac{2}{\sqrt{3}}x^{1/2}-\frac{1}{2\sqrt{3}}x^{3/2} \\
\frac{dy}{dx} &= \frac{1}{\sqrt{3}}x^{-1/2}-\frac{3}{4\sqrt{3}}x^{1/2} \\
1+\frac{dy}{dx}^2 &= 1+\left(\frac{1}{\sqrt{3x}}-\frac{3\sqrt{x}}{4\sqrt{3}}\right)^2 \\
&= 1+\frac{1}{3x}-\frac{1}{2}+\frac{3x}{16} = \frac{1}{2}+\frac{1}{3x}+\frac{3x}{16} \\
&= \frac{9x^2+24x+16}{48x} = \frac{(3x+4)^2}{48x} \\
\sqrt{1+\left(\frac{dy}{dx}\right)^2} &= \sqrt{\frac{(3x+4)^2}{48x}} = \frac{3x+4}{4\sqrt{3}\sqrt{x}} \\
y\sqrt{1+\left(\frac{dy}{dx}\right)^2} &= \left(\frac{2\sqrt{x}}{\sqrt{3}}-\frac{x\sqrt{x}}{2\sqrt{3}}\right)\frac{3x+4}{4\sqrt{3}\sqrt{x}} \\
&= \frac{2\sqrt{x}}{\sqrt{3}}\left(\frac{3x+4}{4\sqrt{3}\sqrt{x}}\right)-\frac{x\sqrt{x}}{2\sqrt{3}}\left(\frac{3x+4}{4\sqrt{3}\sqrt{x}}\right) \\
&= \frac{3x+4}{6}-\frac{3x^2+4x}{24} = \frac{1}{24}(-3x^2+8x+16) \\
S &= 2\pi\int_1^4 y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \frac{2\pi}{24}\int_1^4(-3x^2+8x+16)\,dx \\
&= \frac{\pi}{12}[-x^3+4x^2+16x]_1^4 \\
&= \frac{\pi}{12}[(-64+64+64)-(-1+4+16)] \\
&= \frac{45\pi}{12} = \frac{15\pi}{4}\ \text{units}^2
\end{aligned}
$$

2)

$$
\begin{aligned}
y &= ax, & \frac{dy}{dx} &= a \\
1+\frac{dy}{dx}^2 &= 1+a^2, & \sqrt{1+\left(\frac{dy}{dx}\right)^2} &= \sqrt{1+a^2} \\
y\sqrt{1+\left(\frac{dy}{dx}\right)^2} &= ax\sqrt{1+a^2}
\end{aligned}
$$

$$S = 2\pi \int_0^k y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^k ax\sqrt{1+a^2}\,dx$$
$$= 2\pi a\sqrt{1+a^2}\left[\frac{1}{2}x^2\right]_0^k = \pi a k^2\sqrt{1+a^2} \text{ units}^2$$

3 a)

$$y = (x+16)^{1/2}, \qquad \frac{dy}{dx} = \frac{1}{2}(x+16)^{-1/2}$$
$$1+\frac{dy}{dx}^2 = 1+\left(\frac{1}{2}(x+16)^{-1/2}\right)^2 = 1+\frac{1}{4(x+16)}$$
$$= \frac{4(x+16)+1}{4(x+16)} = \frac{4x+65}{4(x+16)}$$
$$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{4x+65}{2(x+16)}} = \frac{\sqrt{4x+65}}{2\sqrt{x+16}}$$
$$y\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{x+16}\frac{\sqrt{4x+65}}{2\sqrt{x+16}} = \frac{\sqrt{4x+65}}{2}$$
$$S = 2\pi \int_0^6 y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^6 \left(\frac{\sqrt{4x+65}}{2}\right)dx$$
$$= \pi \int_0^6 (4x+65)^{1/2}\,dx = \pi\left[\frac{1}{6}(4x+65)^{3/2}\right]_0^6$$
$$= \pi\left[\frac{1}{6}(89)^{3/2} - \frac{1}{6}(65)^{3/2}\right] = 165.2 \text{ cm}^2 \text{ (4 SF)}$$

3 b)

Cost: $20 \times 165.2 \times 0.03 = £99$ (to the nearest £1)

4)

$$y = \sqrt{x+1} = (x+1)^{1/2}, \qquad \frac{dy}{dx} = \frac{1}{2}(x+1)^{-1/2} = \frac{1}{2\sqrt{x+1}}$$
$$1+\frac{dy}{dx}^2 = 1+\left(\frac{1}{2\sqrt{x+1}}\right)^2 = 1+\frac{1}{4(x+1)}$$
$$= \frac{4(x+1)+1}{4(x+1)} = \frac{4x+5}{4(x+1)}$$
$$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{4x+5}{4(x+1)}}$$
$$y\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{x+1}\sqrt{\frac{4x+5}{4(x+1)}} = \frac{1}{2}\sqrt{4x+5}$$
$$S = 2\pi \int_0^k y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^k \frac{1}{2}\sqrt{4x+5}\,dx$$
$$= \pi \int_0^k (4x+5)^{1/2}\,dx = \pi\left[\frac{1}{6}(4x+5)^{3/2}\right]_0^k$$

$$= \pi \left[\frac{1}{6}(4k+5)^{3/2} - \frac{5\sqrt{5}}{6} \right]$$

But since $\frac{\pi}{6}(13\sqrt{13} - 5\sqrt{5}) = \pi[(4k+5)^{1/2} - \frac{5\sqrt{5}}{6}]$ then

$$\frac{1}{6}(4k+5)^{3/2} = \frac{13\sqrt{13}}{6}$$

$$(4k+5)^3 = 2197 \;\Rightarrow\; 4k+5 = 13$$

$$\therefore k = 2$$

5)

When $y = 0$, so $0 = k - x^2 \Rightarrow x = \sqrt{k}$

$$y = (k-x^2)^{1/2}, \qquad \frac{dy}{dx} = -x(k-x^2)^{-1/2} = -\frac{x}{\sqrt{k-x^2}}$$

$$1 + \frac{dy}{dx}^2 = 1 + \left(-\frac{x}{\sqrt{k-x^2}}\right)^2 = 1 + \frac{x^2}{k-x^2}$$

$$= \frac{k - x^2 + x^2}{k-x^2} = \frac{k}{k-x^2}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{k}{k-x^2}}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{k-x^2}\sqrt{\frac{k}{k-x^2}} = \sqrt{k-x^2}\frac{\sqrt{k}}{\sqrt{k-x^2}} = \sqrt{k}$$

$$S = 2\pi \int_0^{\sqrt{k}} y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = 2\pi \int_0^{\sqrt{k}} \sqrt{k}\, dx$$

$$= 2\pi[\sqrt{k}x]_0^{\sqrt{k}} = 2\pi[k-0] = 2\pi k$$

$$\therefore \quad 2 \times 2\pi k = 4\pi k \text{ units}^2$$

6 a)

$$\int \sqrt{x^2+1}\, dx$$

Let $x = \sinh t \Rightarrow \frac{dx}{dt} = \cosh t \Rightarrow dx = \cosh t\, dt$

$$\therefore \int \sqrt{x^2+1}\, dx = \int \sqrt{\sinh^2 t - 1} \times \cosh t\, dt$$

$$= \int \sqrt{\cosh^2 t} \times \cosh t\, dt$$

$$= \int \cosh t \times \cosh t\, dt = \int \cosh^2 t\, dt$$

$$\cosh 2t \equiv 2\cosh^2 t - 1 \Rightarrow \cosh^2 t \equiv \frac{1}{2}(\cosh 2t + 1)$$

$$\therefore \int \cosh^2 t\, du = \int \left(\frac{1}{2}(\cosh 2t + 1)\right) dt$$

$$= \frac{1}{2}\int (\cosh 2t + 1)\, dt = \frac{1}{2}\left(\frac{1}{2}\sinh 2t + t\right) + c$$

$$= \frac{1}{4}\sinh 2t + \frac{1}{2}t + c$$

$$\sinh 2t = 2\sinh t\cosh t = 2\sinh t \times \sqrt{\sinh^2 t + 1} = 2x\sqrt{x^2+1}$$

$$\therefore \int \sqrt{x^2+1}\,dx = \frac{1}{4} \times 2x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$$
$$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln(x + \sqrt{x^2+1}) + c$$

6 b)

$$y = e^{2x}, \qquad \frac{dy}{dx} = 2e^{2x}$$

$$1 + \frac{dy}{dx}^2 = 1 + (2e^{2x})^2 = 1 + 4e^{4x}, \qquad \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1+4e^{4x}}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = e^{2x}\sqrt{1+4e^{4x}}$$

$$S = 2\pi \int_0^{\ln 2} y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^{\ln 2} e^{2x}\sqrt{1+4e^{4x}}\,dx$$

Let $u = 2e^{2x} \Rightarrow u^2 = 4e^{4x}$ and $\frac{du}{dx} = 4e^{2x}$

When $x = 0, u = 2$, when $x = \ln 2, u = 8$

$$2\pi \int_0^{\ln 2} e^{2x}\sqrt{1+4e^{4x}}\,dx = 2\pi \int_2^8 \frac{1}{4}\sqrt{1+u^2}\,du$$
$$= \frac{\pi}{2}\int_2^8 \sqrt{1+u^2}\,du$$
$$= \frac{\pi}{2}\left[\frac{1}{2}\operatorname{arsinh} u + \frac{1}{4}\sinh(2\operatorname{arsinh} u)\right]_2^8$$
$$= \frac{\pi}{2}\left[\left(\frac{1}{2}\ln(8+\sqrt{65}) + 4\sqrt{65}\right) - \left(\frac{1}{2}\ln(2+\sqrt{5}) + \sqrt{5}\right)\right]$$
$$= 48.2 \text{ units}^2 \text{ (3 SF)}$$

7 a)

$$y = \frac{1}{3}\cos(3x), \qquad \frac{dy}{dx} = -\sin(3x)$$

$$1 + \frac{dy}{dx}^2 = 1 + (-\sin(3x))^2 = 1 + \sin^2(3x)$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1+\sin^2(3x)}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{1}{3}\cos(3x)\sqrt{1+\sin^2(3x)}$$

$$S = 2\pi \int_0^{\pi/6} y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = 2\pi \int_0^{\pi/6} \frac{1}{3}\cos(3x)\sqrt{1+\sin^2(3x)}\,dx$$
$$= \frac{2\pi}{3}\int_0^{\pi/6} \cos(3x)\sqrt{1+\sin^2(3x)}\,dx \quad \text{as req.}$$

7 b)

Let $u = \sin(3x)$ so $\frac{du}{dx} = 3\cos(3x)$

When $x = 0, u = 0$, when $x = \frac{\pi}{6}, u = 1$

$$\frac{2\pi}{3}\int_0^{\pi/6} \cos(3x)\sqrt{1+\sin^2(3x)}\,dx = \frac{2\pi}{3}\int_0^1 \frac{1}{3}\sqrt{1+u^2}\,du$$

Now consider:

$$\int \sqrt{u^2+1}\,dx$$

Let $u = \sinh t \Rightarrow \frac{du}{dt} = \cosh t \Rightarrow du = \cosh t\,dt$

$$\begin{aligned}\therefore \int \sqrt{u^2+1}\,du &= \int \sqrt{\sinh^2 t - 1} \times \cosh t\,dt \\ &= \int \sqrt{\cosh^2 t} \times \cosh t\,dt \\ &= \int \cosh t \times \cosh t\,dt = \int \cosh^2 t\,dt\end{aligned}$$

$$\cosh 2t \equiv 2\cosh^2 t - 1 \Rightarrow \cosh^2 t \equiv \frac{1}{2}(\cosh 2t + 1)$$

$$\begin{aligned}\therefore \int \cosh^2 t\,du &= \int \left(\frac{1}{2}(\cosh 2t + 1)\right)\,dt \\ &= \frac{1}{2}\int (\cosh 2t + 1)\,dt = \frac{1}{2}\left(\frac{1}{2}\sinh 2t + t\right) + c \\ &= \frac{1}{4}\sinh 2t + \frac{1}{2}t + c\end{aligned}$$

$$\sinh 2t = 2\sinh t\cosh t = 2\sinh t \times \sqrt{\sinh^2 t + 1} = 2x\sqrt{x^2+1}$$

$$\begin{aligned}\therefore \int \sqrt{u^2+1}\,du &= \frac{1}{4} \times 2u\sqrt{u^2+1} + \frac{1}{2}\operatorname{arsinh} u + c \\ &= \frac{1}{2}u\sqrt{u^2+1} + \frac{1}{2}\ln(u + \sqrt{u^2+1}) + c\end{aligned}$$

So that

$$\begin{aligned}\frac{2\pi}{3}\int_0^{\pi/6} \cos(3x)\sqrt{1+\sin^2(3x)}\,dx &= \frac{2\pi}{9}\left[\frac{1}{2}u\sqrt{u^2+1} + \frac{1}{2}\ln(u+\sqrt{u^2+1})\right]_0^1 \\ &= \frac{2\pi}{9}\left[\left(\frac{1}{2}\sqrt{2} + \frac{1}{2}\ln(1+\sqrt{2})\right) - (0+0)\right] \\ &= \left(\frac{\pi\sqrt{2}}{9} + \frac{\pi}{9}\ln(1+\sqrt{2})\right)\ \text{units}^2\end{aligned}$$

8 a)

$$y = \frac{\sqrt{7}}{3}(x+1), \qquad \frac{dy}{dx} = \frac{\sqrt{7}}{3}$$

$$1 + \frac{dy}{dx}^2 = 1 + \left(\frac{\sqrt{7}}{3}\right)^2 = \frac{16}{9}, \qquad \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{16}{9}} = \frac{4}{3}$$

$$x\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \frac{4}{3}x$$

$$S = 2\pi\int_0^5 x\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = \frac{8\pi}{3}\int_2^3 x\,dx$$
$$= \frac{8\pi}{3}\left[\frac{1}{2}x^2\right]_2^3 = \frac{8\pi}{3}\left[\frac{9}{2}-2\right] = \frac{20\pi}{3}\text{ units}^2$$

8 b)

$$y = x^2, \qquad \frac{dy}{dx} = 2x$$
$$1+\frac{dy}{dx}^2 = 1+(2x)^2 = 1+4x^2, \qquad \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{1+4x^2}$$
$$x\sqrt{1+\left(\frac{dy}{dx}\right)^2} = x\sqrt{1+4x^2}$$

$$S = 2\pi\int_1^2 x\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi\int_1^2 x\sqrt{1+4x^2}\,dx$$
$$= 2\pi\int_1^2 x(1+4x^2)^{1/2}\,dx = 2\pi\left[\frac{1}{12}(1+4x^2)^{3/2}\right]_1^2$$
$$= 2\pi\left[\frac{17\sqrt{17}}{12}-\frac{5\sqrt{5}}{12}\right] = \frac{\pi}{6}(17\sqrt{17}-5\sqrt{5})\text{ units}^2$$

8 c)

$$y = \frac{2}{3}(x+3)^{3/2}, \qquad \frac{dy}{dx} = (x+3)^{1/2}$$
$$1+\frac{dy}{dx}^2 = 1+((x+3)^{1/2})^2 = 1+x+3 = x+4$$
$$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{x+4}, \qquad x\sqrt{1+\left(\frac{dy}{dx}\right)^2} = x\sqrt{x+4}$$

$$S = 2\pi\int_0^5 x\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi\int_0^5 x\sqrt{x+4}\,dx$$

Let $u = x+4$ so $\frac{du}{dx} = 1$

When $x = 0, u = 4$, when $x = 5, u = 9$

$$2\pi\int_0^5 x\sqrt{x+4}\,dx = 2\pi\int_4^9 (u-4)u^{1/2}\,dx = 2\pi\int_4^9 (u^{3/2}-4u^{1/2})\,dx$$
$$= 2\pi\left[\frac{2}{5}u^{5/2}-\frac{8}{3}u^{3/2}\right]_4^9$$
$$= 2\pi\left[\left(\frac{486}{5}-72\right)-\left(\frac{64}{5}-\frac{64}{3}\right)\right] = \frac{1012\pi}{15}\text{ units}^2$$

8 d)

$$y = \cosh x, \qquad \frac{dy}{dx} = \sinh x$$

$$1 + \frac{dy}{dx}^2 = 1 + (\sinh x)^2 = 1 + \sinh^2 x = \cosh^2 x$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\cosh^2 x}, \qquad x\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x\cosh x$$

$$S = 2\pi \int_0^{\ln 2} x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = 2\pi \int_0^{\ln 2} x\cosh x\, dx$$

Use Integration by Parts:

$$u = x, \qquad \frac{dv}{dx} = \cosh x, \qquad \frac{du}{dx} = 1, \qquad v = \sinh x$$

$$\therefore \int x\cosh x\, dx = x\sinh x - \int \sinh x\, dx = x\sinh x - \cosh x + c$$

So that:

$$2\pi \int_0^{\ln 2} x\cosh x\, dx = 2\pi[x\sinh x - \cosh x]_0^{\ln 2}$$

$$= 2\pi\left[\left(\frac{3}{4}\ln 2 - \frac{5}{4}\right) - (0 - 1)\right] = \frac{1}{2}\pi(3\ln 2 - 1) \text{ units}^2$$

9)

$$y = ax, \qquad \frac{dy}{dx} = a$$

$$1 + \frac{dy}{dx}^2 = 1 + a^2, \qquad \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + a^2}$$

$$y\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x\sqrt{1 + a^2}$$

$$S = 2\pi \int_0^k x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = 2\pi \int_0^k x\sqrt{1 + a^2}\, dx$$

$$= 2\pi\sqrt{1 + a^2}\left[\frac{1}{2}x^2\right]_0^k = \pi k^2\sqrt{1 + a^2} \text{ units}^2$$

10)

$$y = \operatorname{arcosh} x \;\Rightarrow\; \cosh y = x$$

$$\frac{dx}{dy} = \sinh y \quad \therefore \; \frac{dy}{dx} = \frac{1}{\sinh y} = \frac{1}{\sqrt{x^2 - 1}}$$

$$1 + \frac{dy}{dx}^2 = 1 + \left(\frac{1}{\sqrt{x^2 - 1}}\right)^2 = 1 + \frac{1}{x^2 - 1}$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{1}{x^2 - 1}} = \sqrt{\frac{x^2}{x^2 - 1}}$$

$$x\sqrt{1+\left(\frac{dy}{dx}\right)^2} = x\sqrt{\frac{x^2}{x^2-1}} = \frac{x^2}{\sqrt{x^2-1}}$$

$$S = 2\pi\int_1^2 x\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx = 2\pi\lim_{a\to 1^+}\int_a^2 \frac{x^2}{\sqrt{x^2-1}}\,dx$$

Let $x = \cosh u$ so $\frac{dx}{du} = \sinh u\, d$

So $x^2 - 1 = \cosh^2 u - 1 = \sinh^2 u$

As $a \to 1^+, u = 0$

when $x = 2, u = \operatorname{arcosh} 2$

$$\begin{aligned}
2\pi\lim_{a\to 1^+}\int_a^2 \frac{x^2}{\sqrt{x^2-1}}\,dx &= 2\pi\lim_{a\to 1^+}\int_{\operatorname{arcosh} a}^{\operatorname{arcosh} 2} \frac{\cosh^2 u}{\sqrt{\sinh^2 u}}\sinh u\,du \\
&= 2\pi\int_0^{\operatorname{arcosh} 2}\cosh^2 u\,du \\
&= 2\pi\int_0^{\operatorname{arcosh} 2}\left(\frac{1}{2}+\frac{1}{2}\cosh 2u\right)du \\
&= 2\pi\left[\frac{1}{2}u+\frac{1}{4}\sinh 2u\right]_0^{\operatorname{arcosh} 2} \\
&= 2\pi\left[\left(\frac{1}{2}\operatorname{arcosh} 2+\sqrt{3}\right)-(0+0)\right] \\
&= 2\pi\left(\frac{1}{2}\ln(2+\sqrt{3})+\sqrt{3}\right) \\
&= (2\pi\sqrt{3}+\pi\ln(2+\sqrt{3}))\text{ units}^2 \quad \text{as req.}
\end{aligned}$$

11 a)

$$\frac{dx}{dt} = 6 \Rightarrow \frac{dx}{dt}^2 = 36, \qquad \frac{dy}{dt} = 2 \Rightarrow \frac{dy}{dt}^2 = 4$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 40 \Rightarrow \sqrt{40} = 2\sqrt{10}$$

$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = 2t\times 2\sqrt{10} = 4\sqrt{10}t$$

$$\begin{aligned}
2\pi\int_0^3 y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt &= 2\pi\int_0^3 4\sqrt{10}t\,dt \\
&= 2\pi[2\sqrt{10}t^2]_0^3 = 36\sqrt{10}\pi\text{ units}^2
\end{aligned}$$

11 b)

$$\frac{dx}{dt} = 2t \Rightarrow \frac{dx}{dt}^2 = 4t^2$$

$$\frac{dy}{dt} = 4t(4t^2+1)^{-1/2} = \frac{4t}{\sqrt{4t^2+1}}$$

$$\Rightarrow \frac{dy}{dt}^2 = \left(\frac{4t}{\sqrt{4t^2+1}}\right)^2 = \frac{16t^2}{4t^2+1}$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 4t^2 + \frac{16t^2}{4t^2+1} = \frac{4t^2(4t^2+1)+16t^2}{4t^2+1}$$

$$= \frac{4t^2(4t^2+5)}{4t^2+1}$$

$$\Rightarrow \sqrt{\frac{4t^2(4t^2+5)}{4t^2+1}} = 2t\sqrt{\frac{4t^2+5}{4t^2+1}}$$

$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = \sqrt{4t^2+1}\times 2t\sqrt{\frac{4t^2+5}{4t^2+1}} = 2t\sqrt{4t^2+5}$$

$$2\pi\int_0^1 y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt = 2\pi\int_0^1 2t\sqrt{4t^2+5}\,dt = \pi\int_0^1 4t(4t^2+5)^{1/2}\,dt$$

$$= \pi\left[\frac{1}{3}(4t^2+5)^{3/2}\right]_0^1 = \pi\left(9-\frac{5\sqrt{5}}{3}\right) \text{ units}^2$$

12 a)

$$\frac{dx}{dt} = 8t \Rightarrow \frac{dx}{dt}^2 = 64t^2, \qquad \frac{dy}{dt} = 2t \Rightarrow \frac{dy}{dt}^2 = 4t^2$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 64t^2+4t^2 = 68t^2 \Rightarrow \sqrt{68t^2} = 2\sqrt{17}t$$

$$\text{So } x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = 4t^2\times 2\sqrt{17}t = 8\sqrt{17}t^3$$

$$2\pi\int_1^2 x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt = 2\pi\int_1^2 8\sqrt{17}t^3\,dt$$

$$= 2\pi[2\sqrt{17}t^4]_1^2 = 2\pi(32\sqrt{17}-2\sqrt{17})$$

$$= 60\sqrt{17}\pi \text{ units}^2$$

12 b)

$$\frac{dx}{dt} = 2\sin t\cos t = \sin(2t) \Rightarrow \frac{dx}{dt}^2 = \sin^2(2t)$$

$$\frac{dy}{dt} = 2\cos t\sin t = \sin(2t) \Rightarrow \frac{dy}{dt}^2 = \sin^2(2t)$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = \sin^2(2t)+\sin^2(2t) \Rightarrow \sqrt{2\sin^2(2t)} = \sqrt{2}\sin(2t)$$

$$\text{So } x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2} = \sin^2 t\times\sqrt{2}\sin(2t) = 2\sqrt{2}\sin^3 t\cos t$$

$$2\pi\int_0^{\pi/2} x\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt = 2\pi\int_0^{\pi/2} 2\sqrt{2}\sin^3 t\cos t\,dt$$

$$= 2\pi\left[\frac{\sqrt{2}}{2}\sin^4 t\right]_0^{\pi/2} = 2\pi\left(\frac{\sqrt{2}}{2}-0\right) = \sqrt{2}\pi \text{ units}^2$$

13)

$$\frac{dx}{d\theta} = -r\sin\theta \Rightarrow \left(\frac{dx}{d\theta}\right)^2 = r^2\sin^2\theta$$

$$\frac{dy}{d\theta} = r\cos\theta \Rightarrow \left(\frac{dy}{d\theta}\right)^2 = r^2\cos^2\theta$$

$$\therefore \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = r^2\sin^2\theta + r^2\cos^2\theta$$

$$= r^2(\sin^2\theta + \cos^2\theta) = r^2 \Rightarrow \sqrt{r^2} = r$$

$$\text{So } y\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = r\sin\theta \times r = r^2\sin\theta$$

$$2\pi\int_0^{\pi} y\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\,d\theta = 2\pi\int_0^{\pi} r^2\sin\theta\,d\theta$$

$$= 2\pi[-r^2\cos\theta]_0^{\pi} = 2\pi(r^2 + r^2) = 4\pi r^2 \text{ units}^2 \quad \text{as req.}$$

14 a)

$$\frac{dx}{dt} = -\tanh t\,\text{sech}\,t \Rightarrow \frac{dx}{dt}^2 = (-\tanh t\,\text{sech}\,t)^2 = \tanh^2 t\,\text{sech}^2 t$$

$$\frac{dy}{dt} = \text{sech}^2 t \Rightarrow \frac{dy}{dt}^2 = (\text{sech}^2 t)^2 = \text{sech}^4 t$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = \tanh^2 t\,\text{sech}^2 t + \text{sech}^4 t$$

$$= \text{sech}^2 t(\tanh^2 t + \text{sech}^2 t) = \text{sech}^2 t$$

14 b)

$$\sqrt{\text{sech}^2 t} = \text{sech}\,t$$

$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \tan t \times \text{sech}\,t = \tanh t\,\text{sech}\,t$$

$$2\pi\int_0^{\ln 5} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt = 2\pi\int_0^{\ln 5} \tanh t\,\text{sech}\,t\,dt$$

$$= 2\pi[-\text{sech}\,t]_0^{\ln 5} = 2\pi\left(-\frac{5}{13} - -1\right) = \frac{16\pi}{13} \text{ units}^2$$

15 a)

$$\frac{dx}{dt} = 6(1+t)^{1/2} \Rightarrow \frac{dx}{dt}^2 = (6(1+t)^{1/2})^2 = 36(1+t)$$

$$\frac{dy}{dt} = -6(1-t)^{1/2} \Rightarrow \frac{dy}{dt}^2 = (-6(1-t)^{1/2})^2 = 36(1-t)$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 36(1+t) + 36(1-t) = 72 \Rightarrow \sqrt{72} = 6\sqrt{2}$$

$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 4(1-t)^{3/2} \times 6\sqrt{2} = 24\sqrt{2}(1-t)^{3/2}$$

$$2\pi\int_0^{0.8} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt = 2\pi\int_0^{0.8} (24\sqrt{2}(1-t)^{3/2})\,dt$$

$$= 2\pi \left[-\frac{48}{5}\sqrt{2}(1-t)^{5/2} \right]_0^{0.8}$$
$$= 2\pi \left(-\frac{48}{5}\sqrt{2}(0.2)^{5/2} - -\frac{48}{5}\sqrt{2} \right) = 83.8 \text{ cm}^2 \text{ (3 SF)}$$

15 b)

$15 \times 83.77740\ldots \times 0.01 = £25$ (nearest £1)

16)

$$\frac{dx}{dt} = -3k \sin t \cos^2 t \Rightarrow \frac{dx}{dt}^2 = (-3k \sin t \cos^2 t)^2 = 9k^2 \sin^2 t \cos^4 t$$
$$\frac{dy}{dt} = 3k \cos t \sin^2 t \Rightarrow \frac{dy}{dt}^2 = (3k \cos t \sin^2 t)^2 = 9k^2 \cos^2 t \sin^4 t$$
$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 9k^2 \sin^2 t \cos^4 t + 9k^2 \cos^2 t \sin^4 t$$
$$= 9k^2 \sin^2 t \cos^2 t (\cos^2 t + \sin^2 t) = 9k^2 \sin^2 t \cos^2 t$$
$$\Rightarrow \sqrt{9k^2 \sin^2 t \cos^2 t} = 3k \sin t \cos t$$
$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = k \sin^3 t \times 3k \sin t \cos t = 3k^2 \sin^4 t \cos t$$
$$2\pi \int_{\pi/4}^{\pi/2} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_{\pi/4}^{\pi/2} (3k^2 \sin^4 t \cos t)\, dt$$
$$= 2\pi k^2 \left[\frac{3}{5} \sin^5 t \right]_{\pi/4}^{\pi/2} = 2\pi k^2 \left(\frac{3}{5} - \frac{3\sqrt{2}}{40} \right)$$
$$= \frac{3}{20}\pi k^2 (8 - \sqrt{2}) \text{ units}^2 \quad \text{as req.}$$

17)

$$\frac{dx}{dt} = 2 \Rightarrow \frac{dx}{dt}^2 = 4$$
$$\frac{dy}{dt} = t - \frac{1}{t} \Rightarrow \frac{dy}{dt}^2 = \left(t - \frac{1}{t}\right)^2 = t^2 - 2 + \frac{1}{t^2}$$
$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 4 + t^2 - 2 + \frac{1}{t^2} = t^2 + 2 + \frac{1}{t^2} = \left(t + \frac{1}{t}\right)^2$$
$$\Rightarrow \sqrt{\left(t + \frac{1}{t}\right)^2} = t + \frac{1}{t}$$
$$\text{So } x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = 2t \times \left(t + \frac{1}{t}\right) = 2t^2 + 2$$
$$2\pi \int_1^2 x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt = 2\pi \int_1^2 (2t^2 + 2)\, dt$$
$$= 2\pi \left[\frac{2}{3}t^3 + 2t \right]_1^2 = 2\pi \left(\left(\frac{16}{3} + 4\right) - \left(\frac{2}{3} + 2\right) \right)$$
$$= \frac{40}{3}\pi \text{ units}^2$$

18 a)

Use Integration by Parts:

$$u = \ln x, \qquad \frac{dv}{dx} = \frac{1}{x}, \qquad \frac{du}{dx} = \frac{1}{x}, \qquad v = \ln x$$

$$\therefore \int \frac{\ln x}{x}\,dx = \ln^2 x - \int \frac{\ln x}{x}\,dx$$

So $2\int \frac{\ln x}{x}\,dx = \ln^2 x$

$$\therefore \int \frac{\ln x}{x}\,dx = \frac{1}{2}\ln^2 x + c \quad \text{as req.}$$

18 b)

$$\frac{dx}{dt} = 2 \Rightarrow \frac{dx}{dt}^2 = 4$$

$$\frac{dy}{dt} = t - \frac{1}{t} \Rightarrow \frac{dy}{dt}^2 = \left(t - \frac{1}{t}\right)^2 = t^2 - 2 + \frac{1}{t^2}$$

$$\therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 = 4 + t^2 - 2 + \frac{1}{t^2} = t^2 + 2 + \frac{1}{t^2} = \left(t + \frac{1}{t}\right)^2$$

$$\Rightarrow \sqrt{\left(t + \frac{1}{t}\right)^2} = t + \frac{1}{t}$$

$$\text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \left(\frac{1}{2}t^2 - \ln t\right) \times \left(t + \frac{1}{t}\right)$$

$$= \frac{1}{2}t^3 - t\ln t + \frac{1}{2}t - \frac{\ln t}{t}$$

$$2\pi \int_1^3 y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt = 2\pi \int_1^3 \left(\frac{1}{2}t^3 - t\ln t + \frac{1}{2}t - \frac{\ln t}{t}\right)\,dt$$

Consider $\int t\ln t\,dt$:

Use Integration by Parts:

$$u = \ln t, \qquad \frac{dv}{dx} = t, \qquad \frac{du}{dt} = \frac{1}{t}, \qquad v = \frac{1}{2}t^2$$

$$\therefore \int t\ln t\,dt = \frac{1}{2}t^2\ln t - \frac{1}{2}\int t\,dx = \frac{1}{2}t^2\ln t - \frac{1}{4}t^2 + c$$

So that

$$2\pi \int_1^3 \left(\frac{1}{2}t^3 - t\ln t + \frac{1}{2}t - \frac{\ln t}{t}\right)\,dt = 2\pi\left[\frac{1}{8}t^4 - \frac{1}{2}t^2\ln t + \frac{1}{4}t^2 + \frac{1}{4}t^2 - \frac{1}{2}\ln^2 t\right]_1^3$$

$$= 2\pi\left(\left(\frac{81}{8} - \frac{9}{2}\ln 3 + \frac{9}{2} - \frac{1}{2}\ln^2 3\right) - \left(\frac{1}{8} - 0 + \frac{1}{2} - 0\right)\right)$$

$$= 53.1 \text{ units}^2 \text{ (3 SF)}$$

19)

$$\frac{dx}{d\theta} = e^\theta \sin\theta + e^\theta \cos\theta = e^\theta(\sin\theta + \cos\theta)$$

$$\Rightarrow \left(\frac{dx}{d\theta}\right)^2 = e^{2\theta}(\sin\theta + \cos\theta)^2$$

$$\frac{dy}{d\theta} = e^{\theta}\cos\theta - e^{\theta}\sin\theta = e^{\theta}(\cos\theta - \sin\theta)$$

$$\Rightarrow \left(\frac{dy}{d\theta}\right)^2 = e^{2\theta}(\cos\theta - \sin\theta)^2$$

$$\therefore \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = e^{2\theta}(\sin\theta + \cos\theta)^2 + e^{2\theta}(\cos\theta - \sin\theta)^2$$

$$= e^{2\theta}((\sin\theta + \cos\theta)^2 + (\cos\theta - \sin\theta)^2)$$

$$= e^{2\theta}(\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta + \cos^2\theta - 2\sin\theta\cos\theta + \sin^2\theta)$$

$$= e^{2\theta}(2\sin^2\theta + 2\cos^2\theta) = 2e^{2\theta}$$

$$\Rightarrow \sqrt{2e^{2\theta}} = \sqrt{2}e^{\theta}$$

$$\text{So } y\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = e^{\theta}\cos\theta \times \sqrt{2}e^{\theta} = \sqrt{2}e^{2\theta}\cos\theta$$

$$2\pi\int_0^{\pi/2} y\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\,d\theta = 2\pi\int_0^{\pi/2}(\sqrt{2}e^{2\theta}\cos\theta)\,d\theta$$

Consider $\int e^{2\theta}\cos\theta\,d\theta$:

Use Integration by Parts:

$$u = e^{2\theta}, \quad \frac{dv}{d\theta} = \cos\theta, \quad \frac{du}{d\theta} = 2e^{2\theta}, \quad v = \sin\theta$$

$$\therefore \int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta - 2\int e^{2\theta}\sin\theta\,d\theta$$

Use Integration by Parts:

$$u = e^{2\theta}, \quad \frac{dv}{d\theta} = \sin\theta, \quad \frac{du}{d\theta} = 2e^{2\theta}, \quad v = -\cos\theta$$

$$\therefore \int e^{2\theta}\sin\theta\,d\theta = -e^{2\theta}\cos\theta + 2\int e^{2\theta}\cos\theta\,d\theta$$

So that

$$\int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta - 2\int e^{2\theta}\sin\theta\,d\theta$$

$$= e^{2\theta}\sin\theta - 2\left(-e^{2\theta}\cos\theta + 2\int e^{2\theta}\cos\theta\,d\theta\right)$$

$$\Rightarrow \int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta - 4\int e^{2\theta}\cos\theta\,d\theta$$

$$\Rightarrow 5\int e^{2\theta}\cos\theta\,d\theta = e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta$$

$$\Rightarrow \int e^{2\theta}\cos\theta\,d\theta = \frac{1}{5}e^{2\theta}\sin\theta + \frac{2}{5}e^{2\theta}\cos\theta + c$$

Use this result:

$$2\pi\int_0^{\pi/2}(\sqrt{2}e^{2\theta}\cos\theta)\,d\theta = 2\sqrt{2}\pi\left[\frac{1}{5}e^{2\theta}\sin\theta + \frac{2}{5}e^{2\theta}\cos\theta\right]_0^{\pi/2}$$

$$= 2\sqrt{2}\pi\left(\left(\frac{1}{5}e^{\pi} + 0\right) - \left(0 + \frac{2}{5}\right)\right)$$

$$= \frac{2\sqrt{2}\pi}{5}(e^{\pi} - 2)\ \text{units}^2 \quad \text{as req.}$$

20 a)

Trial solution:

$$\int e^{-2x}(\cos x - \sin x)\,dx = e^{-2x}(A\cos x + B\sin x)$$

$$\begin{aligned}\frac{d}{dx}[e^{-2x}(A\cos x + B\sin x)] &= -2e^{-2x}(A\cos x + B\sin x) + e^{-2x}(-A\sin x + B\cos x)\\ &= e^{-2x}((-2A+B)\cos x + (-A-2B)\sin x)\end{aligned}$$

So $-2A + B = 1$ and $-A - 2B = -1$

Solving simultaneously gives $A = -\frac{1}{5}$ and $B = \frac{3}{5}$

$$\begin{aligned}\therefore \int e^{-2x}(\cos x - \sin x)\,dx &= e^{-2x}\left(-\frac{1}{5}\cos x + \frac{3}{5}\sin x\right)\\ &= -\frac{1}{5}e^{-2x}(\cos x - 3\sin x) \quad \text{as req.}\end{aligned}$$

20 b)

$$\begin{aligned}\frac{dx}{dt} &= e^{-t}(\cos t - \sin t) - e^{-t}(\cos t + \sin t) = -2e^{-t}\sin t\\ \Rightarrow \frac{dx}{dt}^2 &= (-2e^{-t}\sin t)^2 = 4e^{-2t}\sin^2 t\\ \frac{dy}{dt} &= -e^{-t}(\cos t - \sin t) + e^{-t}(-\sin t - \cos t) = -2e^{-t}\cos t\\ \Rightarrow \frac{dy}{dt}^2 &= (-2e^{-t}\cos t)^2 = 4e^{-2t}\cos^2 t\\ \therefore \frac{dx}{dt}^2 + \frac{dy}{dt}^2 &= 4e^{-2t}\sin^2 t + 4e^{-2t}\cos^2 t\\ &= 4e^{-2t}(\sin^2 t + \cos^2 t) = 4e^{-2t}\\ \Rightarrow \sqrt{4e^{-2t}} &= 2e^{-t}\\ \text{So } y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} &= (e^{-t}(\cos t - \sin t)) \times 2e^{-t} = 2e^{-2t}(\cos t - \sin t)\\ 2\pi\int_0^{\pi/2} y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt &= 2\pi\int_0^{\pi/2}(2e^{-2t}\cos t - 2e^{-2t}\sin t)\,dt\end{aligned}$$

Since $\int e^{-2x}(\cos x - \sin x)\,dx = -\frac{1}{5}e^{-2x}(\cos x - 3\sin x)$ then

$$\begin{aligned}2\pi\int_0^{\pi/2}(2e^{-2t}\cos t - 2e^{-2t}\sin t)\,dt &= 4\pi\left[-\frac{1}{5}e^{-2t}(\cos t - 3\sin t)\right]_0^{\pi/2}\\ &= 4\pi\left(\left(\frac{3}{5}e^{-\pi}\right) - \left(-\frac{1}{5}\right)\right) = \frac{4\pi}{5}\left(1 + \frac{3}{e^{\pi}}\right) \text{ units}^2\end{aligned}$$